

# Poisoning the Threshold-Calibration Stage in Federated IoT Anomaly Detection: A Policy-Differentiated Vulnerability Analysis

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**Abstract.** Federated learning (FL) defenses for IoT anomaly detection largely focus on training and aggregation, while the post-training *threshold-calibration stage* lets each client autonomously collect benign calibration scores and derive its detection threshold. This paper characterizes that stage under a score-level contamination model: one compromised client replaces a fraction  $f$  of its calibration buffer with tail-sampled benign reconstruction scores, shifting the threshold without altering model weights, training data, gradients, aggregation, test scores, or labels. Across GLOBAL, LOCAL, and CLUSTER threshold policies on the N-BaIoT nine-device federation, the score-level raising attack achieves a 100% pass rate on the pre-specified materiality gate at all non-zero fractions and degrades victim true-positive rates by 2.4–7.9 percentage points (4.9–7.9 pp at  $f=0.4$ ). The lowering attack raises false-alarm indicators for LOCAL and CLUSTER, increasing cross-client  $\Delta CV(FPR)$  by 0.05–0.06 (at  $f=0.4$ ) under REPLACE-FIXED-BUDGET; the associated  $\Delta TPR > 0$  is threshold displacement, not a security benefit. GLOBAL bounds lowering through  $1/N$  dilution. Policy choice governs both magnitude and propagation: LOCAL confines threshold harm to the compromised client, CLUSTER can transmit small co-cluster perturbations, and GLOBAL dilutes but propagates shifts federation-wide. The results are bounded to N-BaIoT and score substitution; they characterize a calibration-channel vulnerability mechanism, not a complete traffic-generation attack.

**Keywords:** Federated Learning · Anomaly Detection · IoT Security · Poisoning Attack · Threshold Calibration · Vulnerability Analysis

## 1 Introduction

Federated learning (FL) is commonly used for IoT anomaly detection: devices contribute to a shared model without centralizing raw traffic, and personalized detection thresholds let each node adapt to its own traffic distribution [15]. The dominant security concern in federated anomaly detection focuses on the

*training phase*—Byzantine clients injecting adversarial gradients, label-flipping poisoning, or model-inversion attacks [5,4,19]. Aggregation-phase defenses such as FLTrust, Krum, and Trimmed-Mean target exactly this surface [7,6,20]. Recent FL-IDS work extends this landscape with personalized, non-IID-robust, knowledge-distillation, and anti-poisoning designs [3,18,14,9], all targeting the training or aggregation surface rather than the post-training calibration buffer.

**A post-training coverage gap.** The *threshold-calibration stage* is the post-training, pre-deployment step in which each client collects a window of benign traffic, computes local reconstruction scores, and derives its detection threshold at a fixed quantile  $q$ . Training- and aggregation-phase defenses studied in the FL-IDS literature do not monitor this post-training calibration stage: they complete before calibration begins and cannot observe the calibration buffer. If a single client’s benign calibration buffer can be contaminated with scores derived from benign calibration records that are anomalously high (or low) relative to that client’s own distribution, the resulting threshold shifts without any change to model weights, training data, labels, or gradients. No claim is made that a dedicated calibration-stage defense is impossible—only that the training- and aggregation-phase defenses studied in the FL-IDS literature do not cover this stage.

**Study focus.** This paper identifies threshold calibration as an under-analyzed attack surface and characterizes its vulnerability under a score-level contamination model. Concretely, an adversary with access to a single client’s calibration buffer replaces a fraction  $f$  of entries with tail-sampled scores drawn from the same client’s benign distribution (*calibration-stage poisoning*). The evaluation covers three threshold-policy families—GLOBAL (federation-wide mean), LOCAL (per-client), and CLUSTER (cluster-averaged)—across all nine physical-device clients in the N-BaIoT dataset [13] at four injection fractions.

To address the objection—“this is just data poisoning applied to calibration data”—calibration-stage poisoning is structurally distinct along three axes (Table 1): it is *temporally isolated* (strictly after model convergence, bypassing training-phase defenses); it exploits a *phase-specific trust model* (calibration data is collected autonomously with no central validation gate); and its effects are *policy-dependent* (the same injection fraction yields qualitatively different threshold shifts and threshold propagation depending on whether thresholds aggregate globally, locally, or by cluster).

### Contributions.

1. The analysis isolates the post-training threshold-calibration stage as a distinct attack surface: contaminating one client’s benign calibration scores can shift thresholds while model weights, training data, aggregation, test scores, and labels remain fixed (§3).
2. The evaluation presents a controlled vulnerability characterization under score-level calibration contamination across three threshold policies, showing that the score-level raising attack reliably degrades victim detection rates by 2.4 to 7.9 percentage points across non-zero injection fractions (4.9–7.9 pp at  $f=0.4$ ) with 95% bootstrap confidence intervals excluding zero (§5).

3. The results document policy-differentiated behavior: LOCAL gives the largest per-victim threshold shifts with no cross-client threshold propagation; CLUSTER allows small intra-cluster threshold perturbations; GLOBAL dilutes shifts but propagates them fleet-wide; lowering is secondary and conditionally bounded, especially under GLOBAL (§5).

**Scope.** All claims are bounded to the nine-device N-BaIoT setting with autoencoder detectors and FedAvg aggregation, and the poisoning mechanism operates at the score level to isolate the calibration channel; end-to-end traffic-generation realizability is future work (§6). No claim is made about deployability, evasion guarantees, or generalization beyond this setting. Defense design against calibration-stage poisoning is out of scope, though §6 discusses properties a defense would need to satisfy.

## 2 Background and Related Work

**Federated anomaly detection for IoT.** Federated intrusion and anomaly detection for IoT has expanded from simple FedAvg-style training toward personalized, non-IID, knowledge-distillation, clustered, and anti-poisoning designs [3,18,14,9,15]. These studies improve training-time collaboration and robustness metrics, while thresholded autoencoder pipelines still rely on a benign post-training calibration step. The N-BaIoT benchmark [13] provides labeled benign and Mirai/BASHLITE traffic for nine physical devices spanning five device types, establishing a widely used evaluation platform for FL-IDS research.

**Defenses.** Existing defenses target the *training phase*. Byzantine-robust rules, including Krum [6], Trimmed-Mean, and Median [20], filter or downweight outlier gradients before aggregation. FLTrust [7] anchors aggregation to a trusted server dataset. These defenses operate on gradients or model updates; they complete before the threshold-calibration stage begins. Recent FL-IDS defenses extend this training-side view with personalized poisoning defenses, knowledge-distillation verifiers, and probe-guided anti-poisoning pipelines under non-IID IoT data [18,14,9]; the defenses reviewed here do not observe the later client-local calibration buffer.

**Threshold calibration and poisoning.** By construction, a detection threshold at fixed quantile  $q$  is sensitive to the scores admitted to the calibration distribution: tail-high contamination raises the boundary (missed detections); tail-low contamination lowers it (false alarms). Kloft & Laskov [10] study online poisoning in centralized anomaly detection via a continuous training stream, assuming a trusted aggregator or persistent model-update access—both absent from FL’s post-training calibration stage, where calibration data never leaves the client and no server-side sanitization is possible. FL model poisoning [2] also targets the gradient/weight surface. The reviewed literature does not identify prior work that isolates post-training threshold calibration as the attacked channel in federated threshold-based IoT anomaly detection under otherwise-unchanged training, model, aggregation, and test artifacts (Table 1). No claim

**Table 1.** Positioning relative to adjacent poisoning literature. Calibration-stage poisoning differs by acting strictly after training, on the autonomously collected benign calibration buffer, leaving model weights, gradients, training data, and test data unchanged.

| Prior work type                  | Attack surface        | What changes              | Distinction from this work                                           |
|----------------------------------|-----------------------|---------------------------|----------------------------------------------------------------------|
| Training-data poisoning [5,19]   | Training set / labels | Model parameters          | Calibration-stage poisoning leaves training data and model unchanged |
| FL model poisoning [4,2]         | Client gradients      | Model weights             | Needs no gradient access; acts after convergence                     |
| Byzantine aggregation [6,20]     | Aggregation inputs    | Server aggregate          | Does not touch aggregation; harm is post-aggregation                 |
| Centralized AD poisoning [10]    | Central train stream  | Decision boundary         | No trusted sanitizer exists in FL calibration                        |
| Threshold personalization [8,17] | Threshold policy      | Per-client/cluster $\tau$ | Studies utility, not adversarial contamination                       |
| <b>This work</b>                 | Calibration scores    | Threshold $\tau$ only     | Score-level, post-training, single-client                            |

is made to have introduced a new poisoning taxonomy; the contribution is isolating the post-training calibration channel in federated threshold-based anomaly detection and showing policy-dependent victim effects and threshold propagation under a controlled score-level intervention. The closest neighbors treat the calibration/thresholding pipeline as trusted: Laridi et al. [11] propose summary-statistics-based federated thresholding without adversarial contamination, and Sáez-de-Cámara et al. [16] propose a clustered federated IoT anomaly-detection architecture assuming a benign calibration pipeline; calibration-stage poisoning instead treats client-local benign calibration as the attacked channel.

**Threshold policies.** Federated threshold personalization spans GLOBAL, LOCAL, and CLUSTER designs [8,17] (defined precisely in §4); their security implications have not been studied under calibration-stage adversaries.

### 3 Threat Model

**Attack surface.** The threshold-calibration stage occurs after global model convergence and before threshold-dependent detection begins (Fig. 1). Each client  $i$  collects a benign calibration set  $\mathcal{C}_i = \{s_1, \dots, s_{n_i}\}$  of reconstruction scores from its own traffic, then sets its local threshold  $\tau_i = \text{quantile}(\mathcal{C}_i, q)$  at a fixed quantile  $q$  (here  $q=0.95$ ). Depending on the federation’s threshold policy,  $\tau_i$  may be

used directly (LOCAL), averaged across clients (GLOBAL), or averaged within a calibration-score cluster (CLUSTER).

The training-phase and aggregation-phase defenses considered in this study do not monitor or validate the calibration buffer: it is collected autonomously by the client after the aggregation round completes.

**Adversary model.** The adversary controls a single compromised client  $v$ , called the *poisoned client*. Under LOCAL policy, only client  $v$ 's own detection degrades; under GLOBAL policy, all federation members receive the contaminated threshold. The term “victim” refers to client  $v$  itself when measuring per-client detection harm, and “non-victim clients” refers to uninvolved federation members.

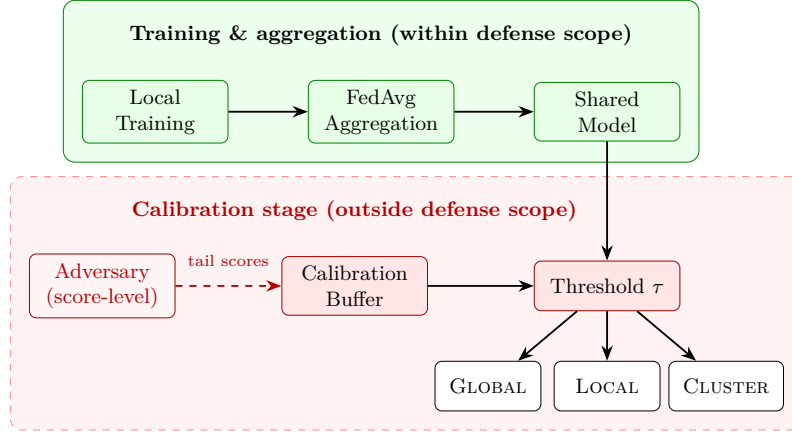
The adversary has *gray-box score-level access*: since the adversary controls the client, it can read and modify only its own local calibration buffer before threshold computation. It cannot modify training data, training labels, test scores, test labels, model weights, gradients, the aggregation procedure, other clients' data, client eligibility, or the threshold-policy definition (GLOBAL/LOCAL/CLUSTER are fixed by the federation). The reservoir is drawn exclusively from the same client's own benign calibration split (§4): attack-labeled samples, test scores, and training scores never enter it, by construction. This is a strictly weaker capability than model-poisoning adversaries, which require gradient injection. The adversary operates at the score level; generating traffic that achieves a target reconstruction score is out of scope (end-to-end realizability is discussed in §6). The REPLACE-FIXED-BUDGET mechanism samples with replacement from the tail reservoir, which at high fractions produces duplicate scores detectable by a uniqueness check (disclosed in §6).

**Attack mechanics.** The evaluation considers two attack objectives:

**Raise.** THRESHOLD-RAISE (primary):  $f=0$  is the unpoisoned clean baseline ( $m=0$ , no replacement); for  $f>0$ , replace  $m = \max(1, \text{round}(fn_v))$  positions in  $\mathcal{C}_v$  with scores drawn with replacement from the top-10% tail of  $\mathcal{C}_v$  (HIGH-SCORE-BENIGN source). All replacement scores are benign-origin reconstruction scores drawn from the victim's own clean calibration distribution; AUROC is invariant by construction (test scores and labels are unchanged under poisoning). A raised threshold  $\tau_v$  increases the detection threshold, suppressing alarms at the victim site.

**Lower.** THRESHOLD-LOWER (secondary, conditional): replace  $m$  positions with scores from the bottom-10% tail (LOW-SCORE-BENIGN). A lowered threshold increases the false-positive rate (alarm burden) and FPR disparity; the resulting  $\Delta\text{TPR}>0$  reflects threshold displacement and is a secondary indicator of operating-point shift. This attack is structurally conditional: random position replacement displaces upper-tail support with probability  $\approx f$ .

**Negative control.** RANDOM-BENIGN: replacement scores are sampled uniformly from the full calibration distribution rather than a tail. Any effect attributable to score contamination rather than position perturbation should vanish in this condition.



**Fig. 1.** Attack surface: threshold calibration occurs after FL training. Green solid boxes mark training/aggregation phases; the dashed red region marks calibration, outside the observed defense scope. Score-level access to one client’s calibration buffer can shift  $\tau$  without modifying model weights, gradients, or labels.

**Scope boundary.** The analysis characterizes the vulnerability at the score level. A complete end-to-end attack would additionally require generating network traffic whose autoencoder reconstruction score lands in the target tail—this is future work (§6). The score-level characterization is a controlled experiment, not a deployed attack: it isolates the calibration-channel vulnerability mechanism. Claims are bounded to the single-client, single-compromise scenario; multi-client collusion is not evaluated.

## 4 Experimental Protocol

### 4.1 Dataset and Federation

The evaluation uses N-BaIoT [13], a labeled IoT network traffic dataset covering nine physical devices spanning five device types (doorbell, baby monitor, security camera, thermostat, webcam). Each device constitutes one federation client; all nine are eligible for calibration. Calibration set sizes range from 2,622 (Ecobee Thermostat) to 35,048 (Philips Baby Monitor) benign samples per device (all  $\geq 100$ , satisfying the eligibility threshold). Each device’s benign stream is partitioned chronologically into a 60% train / 1% gap / 20% calibration / 1% gap / 18% test split; the two gaps avoid temporal leakage. Training and calibration use benign traffic only, and the attack never touches test data, test labels, or training labels. Attack-labeled records are loaded from the N-BaIoT attack folders, scaled with the device scaler fitted on training data, scored as the `test_attack` split, and used only as the positive class for TPR evaluation; they never enter FedAvg training, benign calibration arrays, poisoning reservoirs, threshold recomputation, or injected buffers. The federation trains a per-client autoencoder on benign traffic using FedAvg [12] for feature extraction, then

**Table 2.** Experimental configuration.

| Component                                                        | Configuration                                                           |
|------------------------------------------------------------------|-------------------------------------------------------------------------|
| <i>Autoencoder architecture</i>                                  |                                                                         |
| Input features                                                   | 115 (N-BaIoT statistical traffic features)                              |
| Encoder layers                                                   | 115 $\rightarrow$ 80 $\rightarrow$ 40 $\rightarrow$ 20; decoder mirrors |
| Activation                                                       | ReLU (no batch normalization)                                           |
| Loss                                                             | Mean squared reconstruction error                                       |
| Optimizer                                                        | Adam, lr = 0.001, weight_decay = 0                                      |
| <i>Federated training (FedAvg)</i>                               |                                                                         |
| Local epochs                                                     | 1                                                                       |
| Aggregation                                                      | Sample-count-weighted mean                                              |
| Rounds                                                           | 40 initial, 150 max (conv.: $\Delta\text{loss} \leq 0.005$ over 10)     |
| Batch size                                                       | 256                                                                     |
| <i>Calibration (per-device sizes in released split manifest)</i> |                                                                         |
| Benign samples / device                                          | 2,622 (Ecobee) – 35,048 (Philips)                                       |
| Threshold quantile $q$                                           | 0.95                                                                    |

enters the threshold-calibration stage at a fixed quantile  $q=0.95$ . Table 2 lists the autoencoder architecture, training hyperparameters, and the calibration size range; injection fractions, seeds, and the reservoir are detailed under Attack Parameterization below.

## 4.2 Threshold Policies

Three threshold-policy families are evaluated, covering the main design points in the personalization literature:

### Global.

$$\tau_i^{\text{local}} = \text{quantile}(\mathcal{C}_i, q), \quad \tau^{\text{global}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{local}}.$$

Global assigns the same  $\tau^{\text{global}}$  to every client; a single poisoned local threshold contributes only  $1/N$  of the shared threshold shift.

**Local.** Each client retains  $\tau_i = \tau_i^{\text{local}}$ . Shifts are confined to the compromised client; no cross-client threshold propagation.

**Cluster.** Clients are grouped on a standard-scaled 4-feature fingerprint (mean, std, skew,  $p_{95}$  of  $\mathcal{C}_i$ ) via  $k$ -means++ [1] ( $K=3$ , `random_state`= 42; clustering recomputed after each attack draw so a poisoned fingerprint can change the compromised client’s cluster assignment). Within each cluster, the shared threshold is the cluster mean; the higher seed-to-seed variance in CLUSTER CV(FPR) reflects this cluster-assignment sensitivity. A compromised client may perturb the threshold for co-cluster members (*intra-cluster threshold perturbation*); clients in other clusters are unaffected. Tail injection shifts all four fingerprint components upward, displacing the victim into an effectively singleton cluster whose mean collapses to the victim’s own poisoned

local threshold, producing near-LOCAL victim-level shifts while co-cluster members absorb near-zero perturbation.

### 4.3 Attack Parameterization

For each (policy, attack objective, injection fraction  $f \in \{0.0, 0.1, 0.2, 0.4\}$ ) combination, the protocol runs 10 paired seeds: each seed fixes a training configuration and an independent poisoning draw. A clean copy of the calibration array is preserved; only the poisoned copy is used for threshold computation. The four source-objective pairs are: (1) HIGH-SCORE-BENIGN + THRESHOLD-RAISE (primary raising attack), (2) LOW-SCORE-BENIGN + THRESHOLD-LOWER (primary lowering attack), (3) RANDOM-BENIGN + THRESHOLD-RAISE (objective-matched negative control for the raising attack), and (4) RANDOM-BENIGN paired with THRESHOLD-LOWER (negative control for the lowering attack). The attack is applied to each of the nine clients as victim in turn (single-client scope), yielding 4 source-objective pairs  $\times$  3 threshold policies  $\times$  4 injection fractions  $\times$  9 victim clients  $\times$  10 seeds = 4,320 experimental cells total. Training seed  $i \in \{0, \dots, 9\}$  is paired with poisoning seed  $i+100$  and analysis seed  $i+300$ , separating training, calibration sampling, and bootstrap resampling stochasticity.

### 4.4 Statistical Analysis

**Gate 1 (material shift, primary gate).** Threshold shift  $\Delta\tau_v = \tau_v^{\text{poisoned}} - \tau_v^{\text{clean}}$  is computed per seed. A seed is *Gate-1 significant* when  $|\Delta\tau_v| > \delta\tau_v$  and the sign is consistent with the attack objective, where the materiality threshold is  $\delta\tau_v = \max(0.1 \times \text{IQR}(\mathcal{C}_v^{\text{clean}}), 0.01 \times \text{median}_j(\text{IQR}(\mathcal{C}_j^{\text{clean}})))$ . Gate-1 passes when  $\geq 8/10$  seeds are significant and the victim-majority condition ( $\geq 5/9$  of the nine clients, each taken as victim in turn, are significant) holds.

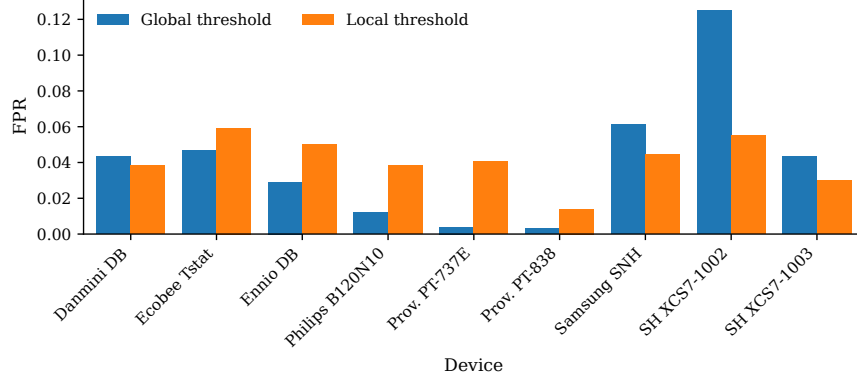
**Gate 2 (directional excess over control).** Bootstrap 95% CI on mean  $\Delta\tau$  excludes zero for the primary strategy but spans zero for the negative control (RANDOM-BENIGN).

**Gate 3 (downstream harm).** Mean victim TPR change,  $\Delta\text{TPR}$ , is directionally consistent (negative for raising, positive for lowering) across  $\geq 8/10$  seeds.

**Confidence intervals.** All CIs are 95% percentile bootstrap over 10 seed-level aggregates, with  $B=10,000$  resamples. Primary claim gates are evaluated over these paired seed-level aggregates; the per-victim-seed cells are descriptive support, not independent units. With ten paired seeds, percentile-bootstrap intervals are approximate uncertainty estimates rather than exact coverage guarantees, and may deviate from nominal at the tails. A two-sided Wilcoxon signed-rank test on ten seed-level mean victim  $\Delta\text{TPR}$  values (raising,  $f=0.4$ ) gives  $W=0$ ,  $p=0.002$  (exact) for all three policies.

**Gate constant justification.** The 10% IQR coefficient for Gate-1 materiality ( $0.1 \times \text{IQR}$ ) exceeds the calibration instability observed in the clean baseline while remaining below the expected tail-injection shift at  $f=0.1$ ; the 1%





**Fig. 2.** Clean-baseline FPR heterogeneity is larger under the shared GLOBAL threshold than under LOCAL calibration (seed 0). All-seed dispersion, including CLUSTER, is reported in Table 3.

cross-client floor ( $0.01 \times \text{median}_j(\text{IQR})$ ) prevents degenerate clients with near-zero score spread from counting as material on arbitrarily small shifts. Both constants were locked from clean-artifact statistics before any poisoned run was executed.

#### 4.5 Isolation Sanity Check

To confirm that the attack is confined to the calibration stage, the protocol verifies: (i) AUROC is invariant under poisoning for all cells, (ii) per-client rank ordering of clean anomaly scores is unchanged, and (iii) model weights before and after calibration are bitwise identical. All three hold in all 4,320 cells.

## 5 Results

### 5.1 Clean-Baseline Policy Profile

Table 3 reports clean-baseline dispersion across all ten seeds. Lower CV(FPR) means more uniform per-client false-positive treatment: LOCAL is lowest ( $0.333 \pm 0.034$ ), GLOBAL highest ( $0.905 \pm 0.039$ ), and CLUSTER intermediate ( $0.615 \pm 0.143$ ) with cluster-assignment sensitivity. Pairwise CV(FPR) gaps are positive with bootstrap CIs excluding zero: G–L 0.572 [0.537, 0.607], G–C 0.291 [0.214, 0.358], and C–L 0.281 [0.201, 0.374] (unrounded per-seed aggregates). Figure 2 shows the per-client FPR distributions for GLOBAL and LOCAL under the clean baseline. This fairness gain is not a uniform accuracy improvement: GLOBAL retains the best P10 Macro-F1 (10th-percentile of per-client macro-F1 across the nine clients; worst-case fairness metric) (0.398 vs. LOCAL 0.335), so Table 3 reports dispersion, detection quality, and mean FPR/TPR rather than a single summary.

**Table 3.** Clean baseline policy profile on N-BaIoT ( $n=10$  seeds, 9 clients). CV: coefficient of variation; BA: balanced accuracy; P10 F1 = P10 Macro-F1; FPR/TPR are client/seed means. Bold = best; \* std  $\approx 0.0002$ .

| Policy  | CV(FPR)                           | CV(TPR)                           | Worst BA          | P10 F1             | FPR   | TPR   |
|---------|-----------------------------------|-----------------------------------|-------------------|--------------------|-------|-------|
| GLOBAL  | 0.905 $\pm$ 0.039                 | <b>0.255<math>\pm</math>0.030</b> | 0.657 $\pm$ 0.008 | 0.398 $\pm$ 0.053  | 0.041 | 0.793 |
| LOCAL   | <b>0.333<math>\pm</math>0.034</b> | 0.296 $\pm$ 0.036                 | 0.655 $\pm$ 0.005 | 0.335 $\pm$ 0.058  | 0.043 | 0.777 |
| CLUSTER | 0.615 $\pm$ 0.143                 | 0.313 $\pm$ 0.009                 | 0.656 $\pm$ 0.004 | 0.299 $\pm$ 0.000* | 0.043 | 0.742 |

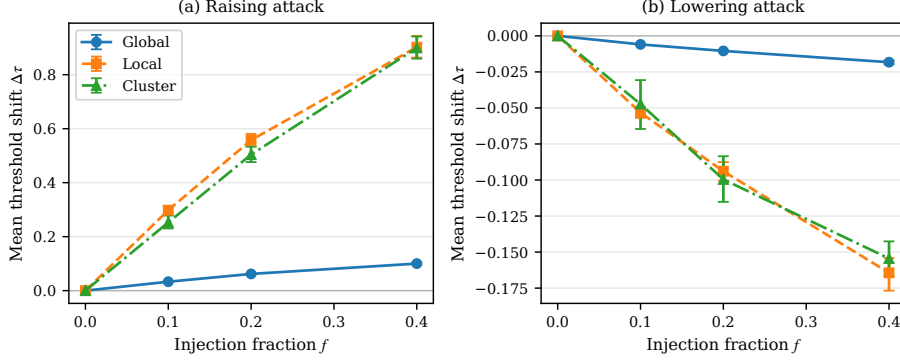
## 5.2 Raising Attack: Primary Results

Table 4 and Fig. 3(a) report threshold shift and downstream detection harm for HIGH-SCORE-BENIGN + THRESHOLD-RAISE.

**Table 4.** Calibration-stage poisoning shifts thresholds under all policies (N-BaIoT,  $n=10$  seeds).  $\Delta\tau$ : mean shift [95 % CI]; G1: Gate-1 seed pass rate;  $\Delta$ TPR is victim detection-rate change. Lowering rows ( $\dagger$ ) use  $\Delta$ CV(FPR) as the primary harm metric; NEG. CONTROL: RANDOM-BENIGN at  $f=0.4$ ; bold=zero G1.

| Policy  | Attack          | $f$ | $\Delta\tau$ [95 % CI]  | G1         | $\Delta$ TPR (pp) | $\Delta$ CV(FPR) |
|---------|-----------------|-----|-------------------------|------------|-------------------|------------------|
| GLOBAL  | Raise           | 0.1 | +0.033 [+0.031, +0.035] | 100 %      | -2.4              | —                |
|         |                 | 0.2 | +0.062 [+0.059, +0.065] | 100 %      | -4.6              | —                |
|         |                 | 0.4 | +0.100 [+0.096, +0.105] | 100 %      | -6.1              | —                |
| LOCAL   | Raise           | 0.1 | +0.298 [+0.281, +0.314] | 100 %      | -6.0              | —                |
|         |                 | 0.2 | +0.558 [+0.535, +0.581] | 100 %      | -7.3              | —                |
|         |                 | 0.4 | +0.902 [+0.864, +0.944] | 100 %      | -7.9              | —                |
| CLUSTER | Raise           | 0.1 | +0.253 [+0.231, +0.277] | 100 %      | -3.9              | —                |
|         |                 | 0.2 | +0.504 [+0.475, +0.532] | 100 %      | -4.9              | —                |
|         |                 | 0.4 | +0.899 [+0.859, +0.941] | 100 %      | -4.9              | —                |
| GLOBAL  | Lower $\dagger$ | 0.1 | -0.006 [-0.006, -0.006] | 60 %       | +0.8              | -0.007           |
|         |                 | 0.2 | -0.010 [-0.011, -0.010] | 100 %      | +1.1              | -0.011           |
|         |                 | 0.4 | -0.018 [-0.020, -0.017] | 100 %      | +2.1              | -0.019           |
| LOCAL   | Lower $\dagger$ | 0.1 | -0.053 [-0.057, -0.049] | 100 %      | +2.8              | -0.002           |
|         |                 | 0.2 | -0.094 [-0.100, -0.088] | 100 %      | +5.9              | +0.005           |
|         |                 | 0.4 | -0.164 [-0.176, -0.154] | 100 %      | +11.8             | +0.053           |
| CLUSTER | Lower $\dagger$ | 0.1 | -0.047 [-0.065, -0.031] | 100 %      | +2.2              | +0.007           |
|         |                 | 0.2 | -0.100 [-0.115, -0.084] | 100 %      | +5.4              | +0.011           |
|         |                 | 0.4 | -0.154 [-0.167, -0.142] | 100 %      | +10.3             | +0.060           |
| GLOBAL  | Neg. Ctrl       | 0.4 | -0.000 [-0.001, +0.001] | <b>0 %</b> | +0.3              | —                |
| LOCAL   | Neg. Ctrl       | 0.4 | -0.002 [-0.009, +0.006] | <b>0 %</b> | +0.9              | —                |
| CLUSTER | Neg. Ctrl       | 0.4 | -0.009 [-0.022, +0.003] | <b>0 %</b> | +0.4              | —                |

**All three policies pass all three gates.** The seed-level Gate-1 pass rate is 100% (10/10 seeds) at every non-zero fraction for all three policies. For LOCAL and CLUSTER, all nine clients are individually significant in every seed.



**Fig. 3.** Mean threshold shift  $\Delta\tau$  vs. injection fraction  $f$  for raising (a) and lowering (b) attacks. Error bars are 95 % bootstrap CIs over 10 seeds; GLOBAL is diluted by  $1/N$ , and RANDOM-BENIGN controls are near-null (Table 4).

For GLOBAL at  $f=0.1$ , two devices (Provision PT-737E and PT-838) are intermittently non-significant (2/10 and 4/10 seeds respectively), but the victim-majority condition ( $\geq 5/9$  per seed) still holds because the remaining seven clients are consistently significant. Bootstrap CIs on  $\Delta\tau$  exclude zero and remain non-overlapping with the RANDOM-BENIGN control, whose CIs span zero, Gate-1 pass rate is 0%, and  $|\Delta\text{TPR}| \leq 1$  pp at every fraction and policy (Table 4 reports representative  $f=0.4$  rows; the artifact contains the full near-null matrix). The targeted effects are therefore attributable to tail contamination, not position perturbation alone.

**Policy magnitude.** Raising-attack magnitude differs by policy. LOCAL and CLUSTER exhibit the largest threshold shifts: at  $f=0.4$ ,  $\Delta\tau = +0.902$  [ $+0.864, +0.944$ ] and  $+0.899$  [ $+0.859, +0.941$ ] respectively. GLOBAL shows a structurally diluted but still significant shift:  $+0.100$  [ $+0.096, +0.105$ ] at  $f=0.4$ . The dilution is expected by construction—GLOBAL’s threshold is the federation mean, so a single-client shift is divided by  $N=9$ —but the residual magnitude ( $\approx 10\%$  of the local shift) still produces measurable downstream harm.

**Downstream harm (Gate 3).** Mean victim  $\Delta\text{TPR}$  is negative and directionally consistent across all seeds for all three policies. At  $f=0.4$ : LOCAL  $-7.9$  pp, CLUSTER  $-4.9$  pp (plateauing from  $f=0.2$ ), GLOBAL  $-6.1$  pp. Absolute clean→poisoned victim TPR at  $f=0.4$  (Table 3 clean column): GLOBAL  $0.793 \rightarrow 0.732$ , LOCAL  $0.777 \rightarrow 0.698$ , CLUSTER  $0.742 \rightarrow 0.693$ . Under GLOBAL, all eight non-victims receive the identical  $\Delta\tau$  (Table 5); their TPR was not separately recomputed, so downstream non-victim impact is inferred, not measured, and heterogeneity ( $\text{CV}(\text{FPR}) = 0.905$ ) means it could differ from the victim’s. LOCAL/CLUSTER non-victim threshold propagation is bounded by construction.

### 5.3 Lowering Attack: Secondary Results (Conditional)

Fig. 3(b) shows threshold shift for LOW-SCORE-BENIGN + THRESHOLD-LOWER. The adversary’s goal is to increase false-alarm burden; the  $\Delta\text{TPR}>0$  below is a threshold-displacement indicator, not a security benefit.

**Local and Cluster succeed; Global is marginal.** LOCAL achieves  $\Delta\tau = -0.164$   $[-0.176, -0.154]$  at  $f=0.4$  with elevated FPR disparity ( $\Delta\text{CV}(\text{FPR}) = +0.053$ ); Gate-3  $\Delta\text{TPR} = +11.8$  pp confirms threshold displacement. CLUSTER similarly achieves  $-0.154$   $[-0.167, -0.142]$  at  $f=0.4$  with  $\Delta\text{CV}(\text{FPR}) = +0.060$  and  $\Delta\text{TPR} = +10.3$  pp. GLOBAL is structurally bounded: at  $f=0.4$ ,  $\Delta\tau = -0.018$   $[-0.020, -0.017]$ , Gate-1 passes only at  $f \geq 0.2$ , and  $\Delta\text{TPR} = +2.1$  pp.

**FPR disparity (operational harm).** The operational harm metric is  $\Delta\text{CV}(\text{FPR})$  (dimensionless):  $+0.053$  (LOCAL, 10/10 seeds) and  $+0.060$  (CLUSTER, 8/10 seeds) at  $f=0.4$ , while GLOBAL shows  $-0.019$  because shared-threshold proportional FPR rise reduces relative disparity. Under a shared threshold,  $\Delta\text{CV}(\text{FPR})$  understates harm: the absolute FPR rises ( $0.041 \rightarrow 0.043$ , Table 3), so the negative value reflects dispersion reduction, not reduced alarm burden. Absolute clean  $\rightarrow$  poisoned FPR at  $f=0.4$  (Table 3 clean column): GLOBAL  $0.041 \rightarrow 0.043$ , LOCAL  $0.043 \rightarrow 0.046$ , CLUSTER  $0.043 \rightarrow 0.049$ .

**Mechanism and conditionality.** Lowering a  $q=0.95$  quantile requires displacing scores in the upper-tail support; random position replacement achieves this with probability  $\approx f$  (40% at  $f=0.4$ ), leaving 60% of upper-tail support intact. The GLOBAL policy compounds this via  $1/N$  dilution. These results are presented as a lower bound under REPLACE-FIXED-BUDGET; targeted upper-tail removal would achieve larger shifts.

### 5.4 Policy-Differentiation Summary

**Table 5.** GLOBAL propagates diluted threshold shifts federation-wide, while LOCAL confines them to the victim ( $f=0.4$ ,  $n=10$  seeds). Blast counts significant non-victim threshold shifts, not measured downstream harm. <sup>†</sup> Shared threshold; <sup>‡</sup> K-means churn, mean non-victim  $\Delta\tau=0.0003$ , std = 0.017.

| Policy  | Victim $\Delta\tau$ | Victim $\Delta\text{TPR}$ (pp) | Non-vic. $\Delta\tau$      | Blast         |
|---------|---------------------|--------------------------------|----------------------------|---------------|
| GLOBAL  | +0.100              | −6.1                           | +0.100 <sup>†</sup>        | $\approx 8/8$ |
| CLUSTER | +0.899              | −4.9                           | $\approx 0.000^{\ddagger}$ | $\approx 6/8$ |
| LOCAL   | +0.902              | −7.9                           | 0.000                      | 0/8           |

Table 5 quantifies the threshold-propagation contrast under the raising attack at  $f=0.4$ . GLOBAL propagates the threshold shift to all nine clients: the non-victim  $\Delta\tau$  equals the victim  $\Delta\tau$  by construction (shared threshold), and blast encompasses  $\approx 8/8$  non-victims in nearly every seed. CLUSTER achieves near-LOCAL victim-level harm; non-victim threshold changes are small in magnitude (mean  $\Delta\tau \approx 0.0003$ , std = 0.017, driven by K-means cluster reassignment

rather than aggregation), with 6/8 non-victims showing statistically significant but small-magnitude threshold perturbations. The blast count measures threshold significance, not downstream harm: non-victim rates were not recomputed for CLUSTER. The mean non-victim  $\Delta\tau$  ( $\approx 0.0003$ ) is two to three orders of magnitude below the weakest victim shifts ( $f=0.1$ : +0.033 GLOBAL, +0.253 CLUSTER, +0.298 LOCAL; Table 4), so large non-victim  $\Delta\text{TPR}$  changes are implausible but unverified—unlike GLOBAL, which transmits  $\Delta\tau$  to every non-victim by construction (also inferred, not measured). LOCAL yields the highest per-victim shift with zero non-victim blast by policy design.

Policy design trades isolated per-victim exposure (LOCAL) against diluted fleet-wide propagation (GLOBAL); CLUSTER carries churn-level threshold perturbations without direct aggregation propagation.

## 6 Discussion and Limitations

**Score-level proxy and detectability.** The experiment operates at the score level: reconstruction scores in the calibration buffer are replaced programmatically rather than through generated traffic. A complete end-to-end attack would additionally require generating network packets whose autoencoder reconstruction error lands in the target tail of the victim’s score distribution—a non-trivial requirement spanning feature engineering, the encoder-decoder nonlinearity, and physical network behavior. This work is a *vulnerability-surface characterization*, not a deployed attack; traffic-level realizability is future work. A related implementation limitation: with-replacement sampling from the tail reservoir at  $f=0.4$  produces duplicate scores in the calibration buffer—for the smallest per-client calibration set ( $n_v \approx 2,622$ ), on the order of  $10^3$  expected collision pairs, easily flagged by a uniqueness check; evading this requires score perturbation at the cost of leaving the exact tail boundary. Detectability remains an open problem.

**GLOBAL threshold propagation, lowering bounds, and policy implications.** The  $1/N$ -diluted GLOBAL shift is definitional; what is empirical is that the residual magnitude ( $\approx 10\%$  of the local shift,  $\Delta\text{TPR} = -6.1$  pp at  $f=0.4$ ) remains measurable above the seed-to-seed variance. The Local/Cluster lowering results are a *lower bound* under REPLACE-FIXED-BUDGET (§5): adversaries with access to the buffer’s rank structure could achieve larger threshold displacements and alarm-burden effects. These results reveal a structural trade-off in policy design: LOCAL maximizes per-client isolation but maximizes per-victim exposure, whereas GLOBAL limits per-victim shift via averaging but propagates residual threshold shifts federation-wide; defense designers must choose which risk to bound.

**Scope limitations.** Claims are bounded to N-BaIoT (9 clients), the autoencoder architecture, FedAvg, and  $q=0.95$ ; generalization to other datasets, larger federations, and quantiles is open. The experiment assumes a single compromised client; multi-client collusion is a direct extension. Generalizability to other anomaly detectors, FL algorithms, and federation profiles is not established. The  $n=10$  seed bootstrap CI coverage may differ from nominal at the tails. Defense

design for the calibration channel (threshold validation, calibration-set integrity constraints) is out of scope and merits independent study.

## 7 Conclusion

This paper characterized the post-training threshold-calibration stage in federated IoT anomaly detection as an attack surface that the studied training-phase and aggregation-phase defenses do not address. Under a score-level contamination model on the N-BaIoT nine-device federation (FedAvg autoencoder,  $q=0.95$ , single compromised client), the raising attack degrades victim detection rates by 2.4–7.9 pp across non-zero injection fractions (4.9–7.9 pp at  $f=0.4$ ) with 100 % Gate-1 pass rates and 95 % bootstrap CIs excluding zero; these intervals are descriptive uncertainty estimates over ten paired seeds. The lowering attack shifts thresholds downward, raising the operating-point-displacement indicator  $\Delta\text{TPR}$  by 10–12 pp—not a security benefit—and the primary alarm-burden metric  $\Delta\text{CV}(\text{FPR})$  by 0.05–0.06 (at  $f=0.4$ , lower bound under REPLACE-FIXED-BUDGET) for LOCAL and CLUSTER. The policy ordering—per-victim exposure maximized under LOCAL, diluted but federation-wide threshold propagation under GLOBAL, small intra-cluster threshold perturbations under CLUSTER—implies calibration-stage hardening should be matched to the threshold policy, not only the training-phase defense. These findings isolate a necessary calibration-channel vulnerability mechanism, not a complete traffic-generation attack. Future work should address traffic-level realizability, calibration-stage integrity defenses, and generalization across detector architectures and datasets.

**Reproducibility.** The released artifact bundle is archived on Zenodo under DOI 10.5281/zenodo.21196568. It contains the experiment configuration, paired training/poisoning/analysis seeds, split manifest with per-device calibration sizes, per-cell results, negative-control records, isolation sanity checks, and figure/table scripts. Model checkpoints, hyperparameters, and analysis gates were fixed before poisoned runs; no poisoned metric was used for checkpoint, hyperparameter, or gate selection.

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## References

1. Arthur, D., Vassilvitskii, S.: k-means++: The advantages of careful seeding. In: SODA. pp. 1027–1035 (2007), <https://dl.acm.org/doi/10.5555/1283383.1283494>
2. Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: AISTATS. pp. 2938–2948. PMLR (2020), <https://proceedings.mlr.press/v108/bagdasaryan20a.html>
3. Belenguer, A., Pascual, J.A., Navaridas, J.: A review of federated learning applications in intrusion detection systems. *Computer Networks* **258**, 111023 (2025). doi:10.1016/j.comnet.2024.111023
4. Bhagoji, A.N., Chakraborty, S., Mittal, P., Calo, S.: Analyzing federated learning through an adversarial lens. In: ICML. pp. 634–643. PMLR (2019), <https://proceedings.mlr.press/v97/bhagoji19a.html>

5. Biggio, B., Nelson, B., Laskov, P.: Poisoning attacks against support vector machines. In: ICML. pp. 1467–1474 (2012), <https://icml.cc/2012/papers/880.pdf>
6. Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: NeurIPS. vol. 30, pp. 119–129 (2017), <https://papers.nips.cc/paper/6617-machine-learning-with-adversaries-byzantine-tolerant-gradient-descent>
7. Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: NDSS (2021). doi:10.14722/ndss.2021.24434
8. Ghosh, A., Chung, J., Yin, D., Ramchandran, K.: An efficient framework for clustered federated learning. In: NeurIPS. vol. 33, pp. 19586–19597 (2020), <https://proceedings.neurips.cc/paper/2020/hash/e32cc80bf07915058ce90722ee17bb71-Abstract.html>
9. Khang, T.T., Duc, T.H., Huynh, D.V., Pham, V.H., Duy, P.T.: P4P: A probe-guided anti-poisoning defense for federated learning-based intrusion detection in IoT networks under non-IID data. Journal of Network and Computer Applications **251**, 104502 (2026). doi:10.1016/j.jnca.2026.104502
10. Kloft, M., Laskov, P.: Security analysis of online centroid anomaly detection. Journal of Machine Learning Research **13**(1), 3681–3724 (2012), <https://www.jmlr.org/papers/v13/kloft12b.html>
11. Laridi, S., Palmer, G., Tam, K.M.M.: Enhanced federated anomaly detection through autoencoders using summary statistics-based thresholding. Scientific Reports **14**(1), 26704 (2024). doi:10.1038/s41598-024-76961-2
12. McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: AISTATS. pp. 1273–1282. PMLR (2017), <https://proceedings.mlr.press/v54/mcmahan17a.html>
13. Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., Elovici, Y.: N-BaIoT—network-based detection of IoT botnet attacks using deep autoencoders. IEEE Pervasive Computing **17**(3), 12–22 (2018). doi:10.1109/MPRV.2018.03367731
14. Quyen, N.H., Duy, P.T., Nguyen, N.T., Khoa, N.H., Pham, V.H.: FedKD-IDS: A robust intrusion detection system using knowledge distillation-based semi-supervised federated learning and anti-poisoning attack mechanism. Information Fusion **117**, 102807 (2025). doi:10.1016/j.inffus.2024.102807
15. Rey, V., Sánchez Sánchez, P.M., Huertas Celdrán, A., Bovet, G.: Federated learning for malware detection in IoT devices. Computer Networks **204**, 108693 (2022). doi:10.1016/j.comnet.2021.108693
16. Sáez-de-Cámara, X., Flores, J.L., Arellano, C., Urbietta, A., Zurutuza, U.: Clustered federated learning architecture for network anomaly detection in large scale heterogeneous IoT networks. Computers & Security **131**, 103299 (2023). doi:10.1016/j.cose.2023.103299
17. Sattler, F., Müller, K.R., Samek, W.: Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. IEEE Transactions on Neural Networks and Learning Systems **32**(8), 3710–3722 (2021). doi:10.1109/TNNLS.2020.3015958
18. Thein, T.T., Shiraishi, Y., Morii, M.: Personalized federated learning-based intrusion detection system: Poisoning attack and defense. Future Generation Computer Systems **153**, 182–192 (2024). doi:10.1016/j.future.2023.10.005
19. Tolpegin, V., Truex, S., Gursay, M.E., Liu, L.: Data poisoning attacks against federated learning systems. In: ESORICS. pp. 480–501. Springer (2020). doi:10.1007/978-3-030-58951-6\_24
20. Yin, D., Chen, Y., Kannan, R., Bartlett, P.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: ICML. pp. 5650–5659. PMLR (2018), <https://proceedings.mlr.press/v80/yin18a.html>